Corrected Coordinate-Frame Replay of the Lunar Prospector Geometry-Only Identifiability Audit

Experiment 008, ECLIPJ2000 Replay v1

14 July 2026

Abstract

We replayed a prospectively frozen geometry-only audit of global lunar surface-potential reconstruction after discovering that Lunar Prospector (LP) spin-axis right ascension and declination had been interpreted in equatorial J2000 rather than their documented Mean-of-J2000 ecliptic frame. The replay constructs the despun-spacecraft frame from spin and Sun vectors in `ECLIPJ2000`, then applies the SPICE transform from `ECLIPJ2000` to `IAU_MOON`. No fitted potential, fit-quality selection, beam estimate, or spacecraft-potential estimate enters the calculation. The corrected extractor agrees with the production geometry implementation to below 1.1×10^{-12} degrees on a one-day fixture. Corrected footprints differ from the old equatorial interpretation by 67.0 km median and 194.8 km at the 95th percentile, and the connected mission cohort decreases from 399,643 to 389,317 sweeps. Despite this material geometric change, the prospectively frozen decision is unchanged. All 560 independent 24-hour degree-5 operators are algebraically full rank, but all fail physical stability and 25-V recovery gates. Degree 1 becomes recoverable at 72 hours, degrees 1–3 at 168 hours, and the frozen synodic-length classes through degree 10 pass. Whole-mission static fields through degree 20 and shared constant-plus-synodic fields through degree 10 also pass. The corrected replay therefore certifies Outcome C: independent daily degree-5 maps are not recoverable, while specified coarse and shared field classes are conditionally identifiable. Absolute lunar potential remains unresolved because spacecraft potential is confounded with monopole terms.

1 Scientific question and information barrier

LP sampled a narrow moving orbital track. A regularized inversion can always return a smooth global field, but the existence of a fitted map does not show that the observations determine that map. The experiment asks which declared spatial and spatiotemporal field classes are identifiable from the actual sampling geometry before any measured potential values are introduced.

The audit reads only raw Level-1 ER/MAG timing and magnetic-field columns, the attitude product, SPICE kernels, and tracked geometry code. It is prohibited from reading fitted potentials, acceptance flags, beam voltages, spacecraft-potential estimates, or potential-cache scientific columns. Each structurally valid and surface-connected 15-row energy sweep is granted an oracle-known scalar relative potential. This is an optimistic information bound: selecting fewer successful fits cannot add geometric rank or visibility. Conversely, a passing geometry class does not validate the measurement estimator.

The original protocol, basis conventions, window grid, seeds, noise scales, rank thresholds, recovery gates, and decision tree are unchanged. The replay modifies only the coordinate interpretation and adds the old equatorial geometry as an explicit sensitivity. Original results and caches remain immutable in a separate namespace.

2 Coordinate-frame correction

2.1 Documented attitude frame

The LP attitude archive defines spin-axis right ascension and declination in the Mean-of-J2000 ecliptic frame, represented by SPICE as `ECLIPJ2000`. Equatorial `J2000` is a distinct inertial frame. Treating the numerical attitude components as equatorial coordinates rotates the physical spin axis by the J2000 obliquity relation and therefore changes the instrument-to-lunar transformation.

For measurement epoch t, the corrected chain is

$$\text{SCD} \longrightarrow \texttt{ECLIPJ2000} \longrightarrow \texttt{IAU_MOON}. \tag{1}$$

Let $\hat{z}$ be the attitude-derived unit spin vector and $\hat{s}$ the LP-to-Sun unit vector, both expressed in `ECLIPJ2000`. The despun-spacecraft axes are

$$\hat{x} = \frac{\hat{s} - (\hat{s} \cdot \hat{z})\hat{z}}{\|\hat{s} - (\hat{s} \cdot \hat{z})\hat{z}\|}, \tag{2}$$

$$\hat{y} = \hat{z} \times \hat{x}. \tag{3}$$

The matrix $M_{\text{SCD}\to\text{ECLIP}}$ has $[\hat{x}\ \hat{y}\ \hat{z}]$ as columns. SPICE supplies

$$M_{\text{ECLIP}\to\text{MOON}}(t) = \text{pxform}(\texttt{ECLIPJ2000}, \texttt{IAU_MOON}, t), \tag{4}$$

and the composite transform is their matrix product. Ephemeris vectors used directly in lunar coordinates are requested in `IAU_MOON` with `spkpos`; no unnecessary intermediate rotation is applied.

Attitude is evaluated at each measurement time by shortest-arc circular interpolation in right ascension and linear interpolation in declination. A legacy next-sample lookup is retained only as a sensitivity.

2.2 Independent implementation and provenance checks

On the one-day fixture, 10,056 surface-connected production rows and replay rows have identical validity and polarity masks. Maximum absolute latitude and longitude discrepancies are 1.05×10^{-12} and 4.26×10^{-13} degrees. Thus the vectorized experiment extractor and the production `CoordinateCalculator` independently implement the same corrected contract.

The corrected and old equatorial footprints differ by 67.039 km median, 194.796 km at the 95th percentile, and 432.581 km maximum on common fixture rows. The new replay's `equatorial` arrays are bit-for-bit identical to the old experiment's canonical arrays, while sweep identifiers, UTC rows, structural masks, and the complete frozen input manifest are exactly unchanged. This isolates the replay difference to the intended coordinate interpretation.

3 Observation operator and field metric

One potential observation belongs to one energy sweep, not to each of its 15 energy rows. For spatial real spherical harmonics R_k and temporal factors T_q, the canonical operator row is

$$A_{i,qk} = \frac{1}{n_i} \sum_{j \in i} T_q(t_{ij}) R_k(x_{ij}), \tag{5}$$

where the sum includes connected row footprints. Repeating one fitted scalar as 15 independent observations would create false precision and is evaluated only as a labeled sensitivity.

Table 1: Mission geometry census.

Quantity	Count
Raw files / file failures	560 / 0
Exact 15-row sweeps	559,321
Structurally valid sweeps	551,637
Corrected connected / complete	389,317 / 366,715
Old equatorial connected / complete	399,643 / 376,920
Frame-mixed connected / complete	421,502 / 399,680
Submitted-legacy connected / complete	421,545 / 399,663

The real basis is orthonormal on the sphere and has $p = (L+1)^2$ coefficients through degree L. Its area-mean-square field metric is

$$\|f\|^2_{\text{global}} = \frac{1}{4\pi}\int_{S^2} f(x)^2\, d\Omega = c^T G_x c, \qquad G_x = \frac{I}{4\pi}. \tag{6}$$

For shared constant-plus-synodic factors $T = (1, \cos\omega t, \sin\omega t)$, the metric is $G = G_t \otimes G_x$, with G_t the analytic continuous-time average over the mission interval.

In physical metric coordinates, the decomposed operator is

$$\widetilde{A} = \frac{AG^{-1/2}}{\sqrt{n}} = U\Sigma V^T. \tag{7}$$

Each singular value is therefore a sampled-RMS/global-RMS visibility ratio. This prevents empirical unit-column normalization from scaling away a weak but physically meaningful mode.

Numerical rank uses $\max(n,p)\epsilon_{64}\sigma_{\max}$. A class is unstable if it is rank deficient, has condition number at least 10^6, has deficient effective rank at 10^{-6}, or has weakest physical visibility at most 0.1. At 25 V sweep noise and 100 V truth RMS, a window must have median relative recovery error below 0.25 and 90th-percentile error below 0.5. A class passes only if at least 90% of its eligible windows satisfy the frozen stability and recovery rules.

Direct SVD is used for independent windows. Whole-mission matrices use tall-skinny QR followed by SVD of the final reduced factor, avoiding condition number squaring. Recovery covariance is evaluated with 16,384 deterministic draws per class gate; the original 32 explicit replicates are retained as an independent Monte Carlo check.

4 Results

4.1 Independent daily degree-5 failure

Every one of the 560 daily degree-5 matrices is formally rank 36, but every window is physically unstable and every window fails recovery. The weakest visibility ranges from 1.148×10^{-6} to 3.739×10^{-4}, with median 4.449×10^{-5}. A 100 V global field in such a direction can produce only millivolt-scale sampled variation. The median condition number is 47,266 and seven windows exceed 10^6.

The median predicted unregularized error is 151 times the truth amplitude. Explicit trials yield zero passing windows. At 25 V noise, broadband median relative errors are 144.1 unregularized, 3.46 for TSVD at 10^{-3}, 0.847 for ridge 10^{-3}, 0.862 for ridge 1, and 0.999 for ridge 1000. Reduced error from ridge regularization reflects suppression of unresolved modes rather than recovery of their true amplitudes.

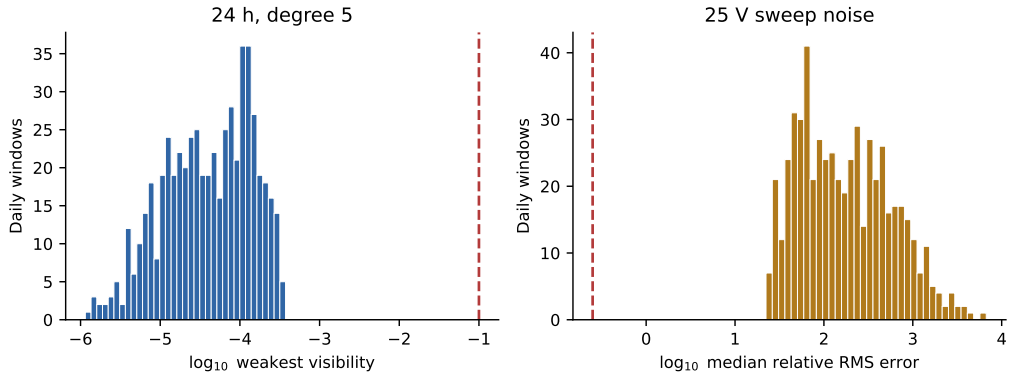


Figure 1: Physical daily degree-5 visibility and recovery failure.

4.2 Resolution boundary

Table 2: Independent-window classes passing the frozen class gate.

Window duration	Passing maximum classes
1 h	none
6 h	none
24 h	none
72 h	$L = 1$
168 h	$L = 1, 2, 3$
708.734112 h	$L = 1, 2, 3, 5, 10$

The set of passing independent classes is exactly the same as in the original run. At 24 hours, degree 1 reaches only 80.7% stable windows. At 72 hours, degree 2 passes recovery but not stability. These cases illustrate why rank, visibility, and recovery were required independently.

Whole-mission static fields pass through degree 20. Their condition numbers range from 1.37 to 3.75; the degree-20 median relative recovery error is 0.00938. Shared constant-plus-synodic fields pass through degree 10. The degree-10 operator has rank 363/363, condition number 3.13, weakest visibility 0.729, and median recovery error 0.00832. Removing the temporal monopoles leaves rank 360/360 and nearly identical diagnostics.

4.3 Sensitivity, blocked validation, and localized fields

Every daily geometry sensitivity has 0% stable and 0% recovery-passing windows. This includes corrected and old equatorial native cohorts, their exact paired cohort, legacy attitude, frame-mixed submitted geometry, midpoint reduction, complete sweeps, solar co-rotation, contrast-only fields, and both row-duplication assumptions. Solar co-rotation raises median visibility to 1.442×10^{-4} but remains nearly 700 times below the 0.1 gate.

All 34 shared-model blocked fits are full rank. Static degree-5 global median errors span 0.263–0.315 V and shared synodic errors span 0.480–0.533 V. Daily degree-5 blocked fits tell the opposite story: contiguous-time holdouts have 29.1 kV median global error but 22.1 V held-out sampled error; longitude holdouts have 17.7 kV global error but 9.7 V sampled error. Local predictive agreement is therefore insufficient evidence for a constrained global map.

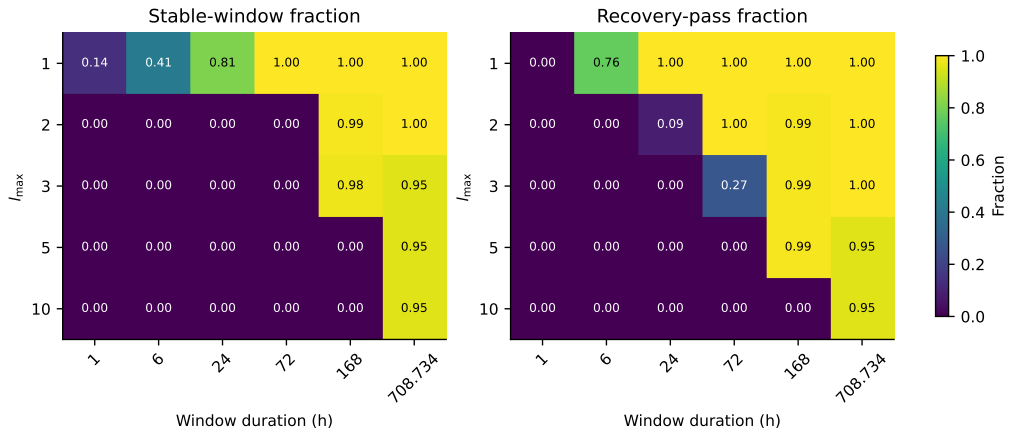


Figure 2: Recoverability boundary versus window duration and spatial degree.

Degree-5 representations of 15–30 degree localized caps have irreducible approximation bias of 80.1 and 41.1 V, respectively, at 100 V truth RMS. Median global reconstruction errors range from 100 V to 15.7 kV and 90th-percentile errors reach 161 kV, even when sampled residuals are small.

5 Comparison with the original run

Table 3: Key original and corrected diagnostics.

Diagnostic	Original	Corrected
Connected canonical sweeps	399,643	389,317
Median daily weakest visibility	4.519×10^{-5}	4.449×10^{-5}
Static degree-20 condition	3.881	3.753
Synodic degree-10 condition	3.205	3.131
Daily stable / recovery windows	0 / 0	0 / 0
Frozen decision	C	C

The coordinate correction is not numerically negligible: it changes footprints by tens to hundreds of kilometers and removes 2.58% of the old connected cohort. Nevertheless, the median daily visibility changes by only -1.56%, the passing class boundary is identical, and corrected mission-integrated controls are marginally better conditioned. Outcome C is therefore robust to the discovered coordinate defect.

Numerical cross-checks support this interpretation. The primary and duplicate grid paths agree exactly in stability and recovery fractions, with median visibility difference 1.25×10^{-17}. The constant block of the temporal cache agrees with the spatial basis to 2.67×10^{-15}, and direct sampled temporal evaluation agrees exactly. Area-weighted grid evaluation gives 99.99746 V for representative nominal 100 V witness fields.

6 Gauge, limitations, and interpretation

The electron measurement is spacecraft-relative. If unknown spacecraft potential $u(t)$ is added to the model, its offset columns duplicate the spatial $\ell = 0$ column and matching temporal monopole columns. The absolute potential problem is exactly rank deficient in those directions. Passing contrast-only classes establish geometric identifiability of spatial variation under their temporal assumptions, not absolute lunar potential.

The 25 V noise scale, Gaussian independence, and oracle sweep scalar are audit assumptions, not measurement validation. Experiment 008 does not establish that loss-cone fitting produces unbiased sweep potentials or calibrated uncertainties. It also cannot exclude temporal model discrepancy: real variability outside a static or single-synodic model can alias into its fitted coefficients.

The scientifically defensible statement is consequently narrower than "no global information" but stronger than a warning about regularization: independent daily degree-5 global fields are not recoverable from LP ER sampling, while substantially longer windows and compact shared temporal models are conditionally identifiable. Those passing operators should be carried forward only after measurement-level experiments establish defensible sweep values, uncertainties, and model-discrepancy treatment.

7 Reproducibility record

The replay uses run identifier `eclipj2000-v1`. Large caches live in an isolated artifact namespace; compact results, figures, freezes, and this supplement live under the tracked replay directory. The principal cache hashes are:

```
geometry: cb47bc11c3bae8fd89eacf1d946db06310f18692c0ff2b5863fcb4dc14eefa62
spatial: 19100c26e64890e9020044bee46ab7e572a6d31af53dcd9c3b606b83c066953e
temporal: e5f7a7345db7e4f53565e5d1037c20e52eaf96f18f32b811c3ccae3b32cf3fdb
```

The original experiment outputs remain unchanged and independently hashable.